

Implement a Chatbot Using Generative AI Techniques

1. Introduction

This project aims to implement a chatbot system using generative AI models. The chatbot mimics human-like conversation based on movie dialogues. It leverages sequence-to-sequence deep learning techniques for generating context-aware responses.

Objectives:

1. Build a chatbot using generative sequence models (LSTM-based).
 2. Use the Cornell Movie-Dialogs Corpus to train the chatbot.
 3. Visualize and analyze dataset statistics.
 4. Evaluate chatbot performance using dialogue testing.
 5. Deploy trained model and test on real-time inputs.
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2. Dataset Description & Preprocessing

2.1. Dataset Overview

- **Dataset Used:** Cornell Movie-Dialogs Corpus.
- The dataset includes over 220,000 conversational exchanges between movie characters from more than 600 movies.
- It contains rich contextual data ideal for training conversational agents.

```
Average input length: 13.314449317738791
Average target length: 13.816299364666811
Max input length: 370
Max target length: 684
```

2.2. Data Preprocessing

The following steps were used to preprocess the data:

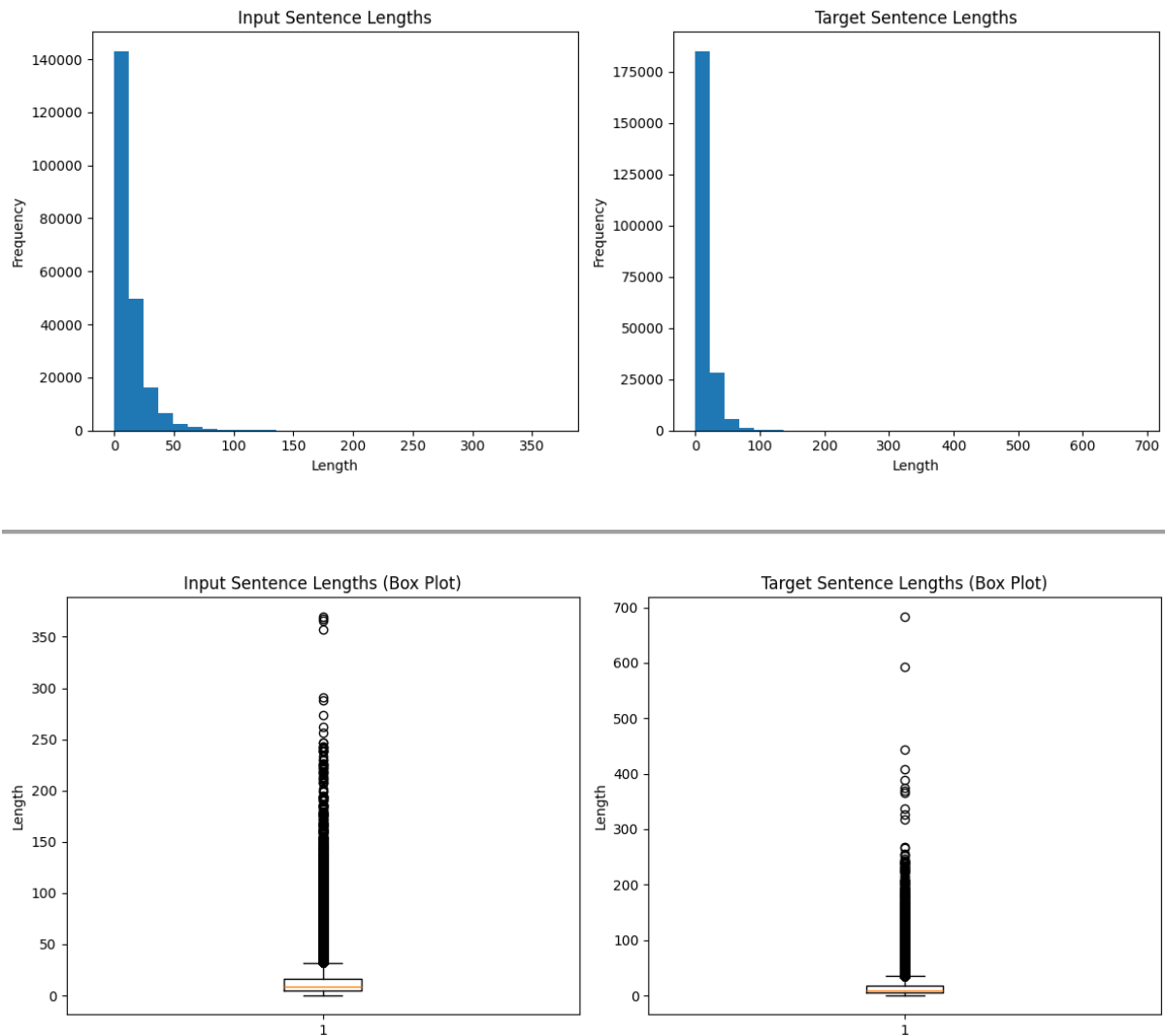
- Cleaned text by removing punctuation, and converting to lowercase.
 - Tokenized using NLTK.
 - Built vocabulary with special tokens: <PAD>, <SOS>, <EOS>, and <UNK>.
 - Encoded questions and answers into numerical sequences.
 - Padded sequences for consistent input lengths.
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3. Exploratory Data Analysis

Sentence Length Distribution:

Visualized using histograms and box plots for both input and target sequences.

Sample Plots:



Observations:

- Average input length: ~13 tokens
 - Average target length: ~13 tokens
 - Max input length: 370
 - Max target length: 684
 - Most conversations are under 50 tokens long.
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4. Model Selection & Implementation

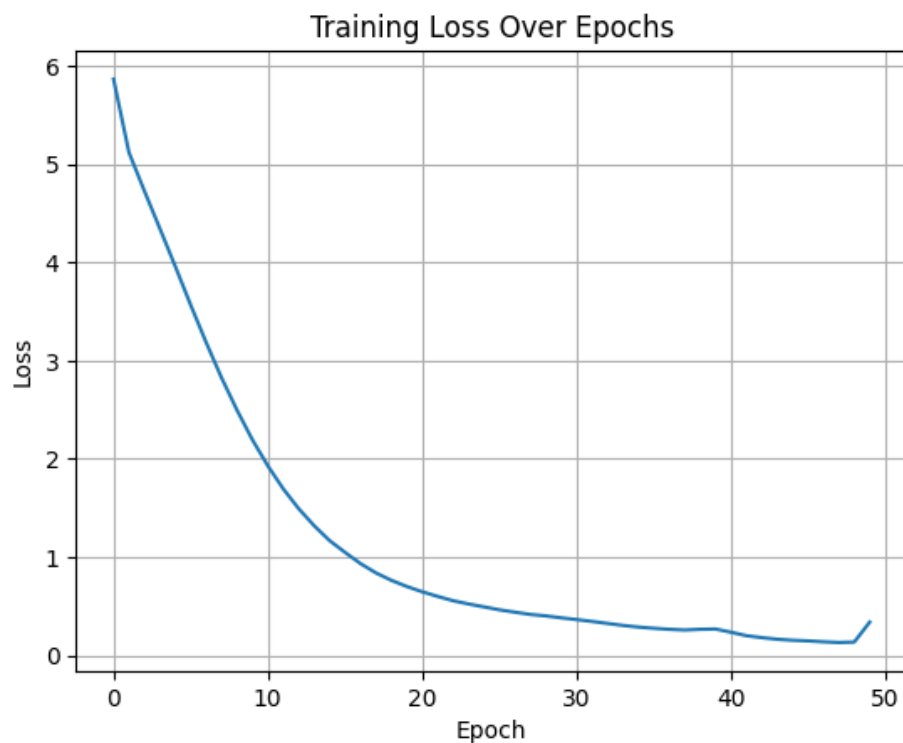
Component	Description
Encoder	Embedding + LSTM encoder to capture input context.
Decoder	LSTM decoder generates a response using the context.
Loss Function	CrossEntropyLoss ignoring <PAD> tokens.
Optimizer	Adam with LR = 0.001.

- Training for 50 epochs.
- Batch size: 32
- Embedding size: 256
- Hidden units: 512

5. Training and Loss Visualization

The model trained on cleaned 10,000 pairs of movie dialogues.

Loss Curve:



Observation:

- Loss steadily decreased and converged after ~45 epochs.
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6. Evaluation and Performance Analysis

Chatbot Testing:

```
do you will be given to any of out
in the course of an evening i think we re in one of a very kind of weather man i
```

User Input	Bot Response
Hello	to be very simple box i'm a huge strong fellow
How are you?	i'm a negro
What's your favourite movie?	your wife may be right...
Tell me a joke.	love changes only and double...
Goodbye	you want to get some food old women...

```
User: Hello
Bot: to be very simple box i m a huge strong fellow
---
User: How are you doing?
Bot: i m a negro
---
User: What's your favorite movie?
Bot: your wife may be right to the fact gracefully out of the way we can go double a fire dead
---
User: Tell me a joke.
Bot: love changes only and double an affair and jack went to that feeling as a fire press out that s
---
User: Goodbye
Bot: you want to get some food old women some old trying to education to hold it away
---
```

Limitations:

- Some responses are grammatically incorrect.
- A few inappropriate outputs due to raw dialogue content.

7. Result Highlights

- Successfully built a functional LSTM-based chatbot.
 - Implemented full training loop, loss tracking, and real-time inference.
 - Integrated data visualizations (sentence lengths, loss curves).
 - Saved deployment model: chatbot_model.pth
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8. Future Improvements

- Integrate attention mechanism for better context retention.
- Filter toxic/biased outputs via data cleansing.
- Use transformer-based models (DialogGPT, BERT2BERT) for improved quality.
- Deploy chatbot on a web app with a user-friendly interface.

9. Conclusion

This project demonstrates how generative AI can be applied to build a chatbot using the Cornell Movie-Dialogs Corpus. The model mimics real dialogues and generates text-based responses in real time. Though basic, it forms the foundation for more advanced conversational agents using transformers.
